

Research Methodology in Public Administration

Proposal Writing and Report Writing

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Data Analysis

Transforming Data into Meaningful Information

Data Collection



Data Preparation



Data Analysis



Interpretation



Findings

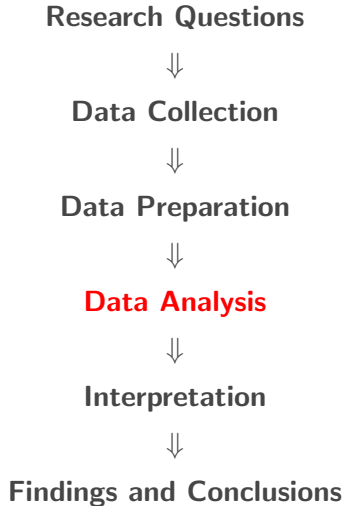
What is Data Analysis?

Data analysis is the systematic process of organizing, processing, examining, and interpreting data in order to answer research questions and achieve research objectives.

Data analysis helps researchers to:

- ▶ Summarize data
- ▶ Identify patterns
- ▶ Examine relationships
- ▶ Test hypotheses
- ▶ Interpret findings
- ▶ Draw conclusions

Data Analysis in the Research Process



Major Types of Data Analysis

Data analysis can broadly be divided into:

Quantitative Analysis

Analysis of numerical data using statistical techniques.

- ▶ Descriptive statistics
- ▶ Inferential statistics
- ▶ Correlation
- ▶ Regression

Qualitative Analysis

Analysis of non-numerical data to understand meanings, experiences, and social phenomena.

- ▶ Coding
- ▶ Categorization
- ▶ Thematic analysis
- ▶ Content analysis

Preparing Data for Analysis

Before analysis, data usually need to be prepared.

Common steps include:

1. Editing
2. Coding
3. Data entry
4. Data cleaning
5. Classification
6. Tabulation

Important

Poor-quality data can lead to misleading findings, regardless of the sophistication of the analysis.

Editing and Coding

Editing

Checking collected data for:

- ▶ Completeness
- ▶ Accuracy
- ▶ Consistency
- ▶ Missing information

Coding

Assigning numbers or labels to responses for easier organization and analysis.

Example:

Male = 1 Female = 2

Levels of Quantitative Data Analysis

Quantitative analysis can be conducted at different levels.

Univariate Analysis



Bivariate Analysis



Multivariate Analysis

- ▶ One variable
- ▶ Two variables
- ▶ Three or more variables

Univariate Analysis

Univariate analysis examines one variable at a time.

It is mainly used to describe the characteristics of the data.

Common techniques include:

- ▶ Frequency
- ▶ Percentage
- ▶ Mean
- ▶ Median
- ▶ Mode
- ▶ Standard deviation

Measures of Central Tendency

Measures of central tendency describe the central or typical value of a dataset.

Mean

The arithmetic average.

$$\bar{X} = \frac{\sum X}{n}$$

Median

The middle value when observations are arranged in order.

Mode

The most frequently occurring value.

Measures of Dispersion

Measures of dispersion show how much the data are spread around the central value.

Common measures include:

- ▶ Range
- ▶ Variance
- ▶ Standard deviation

Example

Two groups may have the same average score but very different levels of variation.

Presenting Quantitative Data

Quantitative data can be presented using:

- ▶ Frequency tables
- ▶ Percentage tables
- ▶ Bar charts
- ▶ Pie charts
- ▶ Histograms
- ▶ Line graphs
- ▶ Scatter plots
- ▶ Statistical tables

Important

The form of presentation should be appropriate for the type of data and the purpose of analysis.

Bivariate Analysis

Bivariate analysis examines the relationship, association, or difference between two variables.

Examples:

- ▶ Education and income
- ▶ Age and political participation
- ▶ Training and employee performance
- ▶ Service quality and citizen satisfaction

Cross-Tabulation

Cross-tabulation is used to examine the relationship between two categorical variables.

	Satisfied	Not Satisfied	Total
Male	40	10	50
Female	35	15	50
Total	75	25	100

The table helps researchers compare patterns between categories.

Correlation Analysis

Correlation analysis measures the direction and strength of the relationship between two variables.

The correlation coefficient generally ranges from:

$$-1 \leq r \leq +1$$

- ▶ +1 = Perfect positive relationship
- ▶ 0 = No linear relationship
- ▶ -1 = Perfect negative relationship

Important

Correlation does not necessarily imply causation.

Inferential Statistics

Inferential statistics are used to make conclusions about a larger population based on data collected from a sample.

Common inferential techniques include:

- ▶ Hypothesis testing
- ▶ t-test
- ▶ Chi-square test
- ▶ Analysis of Variance (ANOVA)
- ▶ Correlation
- ▶ Regression analysis

Hypothesis Testing

A hypothesis is a tentative statement about the relationship or difference between variables.

Two common hypotheses are:

Null Hypothesis (H_0)

There is no significant relationship or difference.

Alternative Hypothesis (H_1)

There is a significant relationship or difference.

Basic Steps in Hypothesis Testing

1. State the null and alternative hypotheses.
2. Select an appropriate statistical test.
3. Determine the significance level.
4. Calculate the test statistic.
5. Examine the p-value.
6. Make a decision regarding the hypothesis.

A commonly used significance level is:

$$\alpha = 0.05$$

Common Statistical Tests

Statistical Test	Purpose
t-test	Compare the means of two groups
ANOVA	Compare the means of three or more groups
Chi-square	Examine association between categorical variables
Correlation	Examine the relationship between variables
Regression	Examine the influence of one or more variables

Regression Analysis

Regression analysis is used to examine how one or more independent variables influence a dependent variable.

Independent Variable(s)



Influence / Predict



Dependent Variable

Example

How do service quality and accessibility influence citizen satisfaction?

Simple Linear Regression

Simple linear regression examines the relationship between:

One Independent Variable
and
One Dependent Variable

The general equation is:

$$Y = a + bX + e$$

Where:

- ▶ Y = Dependent variable
- ▶ X = Independent variable
- ▶ a = Constant
- ▶ b = Regression coefficient

Multiple Regression Analysis

Multiple regression examines the influence of two or more independent variables on one dependent variable.

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3 + e$$

Example

Citizen satisfaction may be influenced by:

- ▶ Service quality
- ▶ Accessibility
- ▶ Staff behavior

Qualitative Data Analysis

Qualitative data analysis focuses on understanding:

- ▶ Meanings
- ▶ Experiences
- ▶ Perceptions
- ▶ Social processes
- ▶ Patterns
- ▶ Context

Qualitative analysis is generally interpretive and iterative.

The Qualitative Data Analysis Process

Interview / Observation / Documents



Transcription and Familiarization



Coding



Categories



Themes



Interpretation

Coding Qualitative Data

Coding is the process of assigning labels to meaningful pieces of qualitative data.

Interview Statement

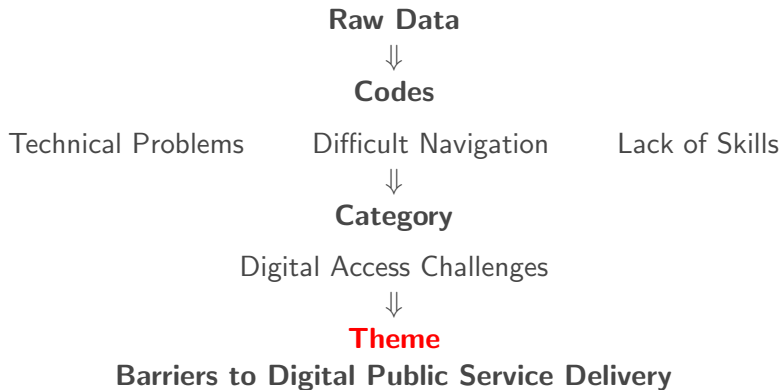
“The online system is useful, but I often face difficulties because the website is difficult to understand.”

Possible codes:

- ▶ Usefulness
- ▶ Technical difficulty
- ▶ Complexity
- ▶ Accessibility challenge

From Codes to Themes

Qualitative analysis generally moves from detailed information toward broader patterns.



Thematic Analysis

Thematic analysis is a method used to identify, analyze, and interpret patterns or themes in qualitative data.

Typical steps include:

1. Familiarize yourself with the data.
2. Generate initial codes.
3. Group related codes.
4. Identify potential themes.
5. Review and refine themes.
6. Interpret and report the findings.

Content Analysis

Content analysis involves systematically examining the content of documents, texts, media, or communication.

It may examine:

- ▶ Words
- ▶ Concepts
- ▶ Themes
- ▶ Categories
- ▶ Frequency of content
- ▶ Patterns in communication

Example

Analyzing government policy documents to identify major policy priorities.

Mixed Methods Data Analysis

Mixed methods research involves analyzing both quantitative and qualitative data.

Quantitative

- ▶ Frequencies
- ▶ Percentages
- ▶ Statistical tests
- ▶ Regression

Qualitative

- ▶ Coding
- ▶ Categories
- ▶ Themes
- ▶ Interpretation

The findings are then integrated to provide a more comprehensive understanding of the research problem.

From Analysis to Interpretation

Analysis and **interpretation** are related but different.

Analysis

What does the data show?

Interpretation

What do the findings mean in relation to the research questions, objectives, theory, and existing literature?

Interpreting Research Findings

When interpreting findings, researchers should consider:

- ▶ Research questions
- ▶ Research objectives
- ▶ Research hypotheses
- ▶ Theoretical framework
- ▶ Previous studies
- ▶ Context of the research

Important

Statistical significance and practical significance are not always the same.

Software for Data Analysis

Different tools can be used depending on the type of data and analysis required.

Quantitative Analysis

- ▶ Excel
- ▶ SPSS
- ▶ Stata
- ▶ R

Qualitative Analysis

- ▶ NVivo
- ▶ ATLAS.ti
- ▶ MAXQDA
- ▶ Manual coding

Data Analysis Plan in a Research Proposal

A research proposal should explain:

1. How the data will be prepared.
2. What type of analysis will be conducted.
3. Which statistical techniques will be used.
4. How qualitative data will be analyzed.
5. Which software or tools will be used.
6. How the findings will answer the research questions.

Example: Quantitative Data Analysis Plan

Example

The collected quantitative data will be edited, coded, and entered into statistical software for analysis. Descriptive statistics, including frequencies, percentages, mean, and standard deviation, will be used to summarize the data. Inferential statistical techniques, including correlation and regression analysis, will be used to examine relationships between the study variables.

Example: Qualitative Data Analysis Plan

Example

The qualitative data collected through semi-structured interviews will be transcribed and systematically reviewed. Relevant sections of the data will be coded and grouped into categories. Themes will then be developed and interpreted in relation to the research questions, objectives, and existing literature.

Linking Research Questions and Analysis

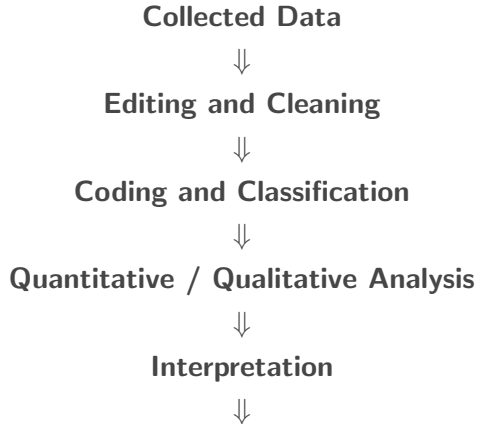
Research Question	Possible Analysis
What are the characteristics of respondents?	Frequency, percentage, mean
Is there a relationship between X and Y?	Correlation / Chi-square
Does X influence Y?	Regression analysis
Are there differences between groups?	t-test / ANOVA
How do participants experience a phenomenon?	Thematic analysis
What themes emerge from policy documents?	Content analysis

Common Mistakes in Data Analysis

Researchers should avoid:

- ▶ Using techniques unrelated to research questions.
- ▶ Selecting inappropriate statistical tests.
- ▶ Ignoring missing or incorrect data.
- ▶ Confusing correlation with causation.
- ▶ Over-interpreting results.
- ▶ Ignoring assumptions of statistical tests.
- ▶ Reporting results without interpretation.

The Overall Data Analysis Process



Findings, Conclusions, and Recommendations

Key Takeaway

Data analysis transforms raw data into research findings.

